to provide new developments, Du Pont began a long range of programme in 1928. Du Pont now shifted from applied research to fundamental research, which primarily aimed to develop knowledge of chemicals, materials and processes.

After many months of research, Dr.Carothers assistant discovered a polymer, which was a clear, heavy molasses when molten and could be drawn out into filament. This was a polymer. It was called 6.6, because there were six carbon atoms per molecule. Later it was called Nylon. It is also known a polyamide.

Nylon is actually a group of related chemical compounds. It is composed of hydrogen, nitrogen, oxygen and carbon in controlled proportions and structural arrangements. Variations in the chemical structures can result in formation of compounds like plastic. The two types of nylon are nylon 6.6 and nylon 6.

3.4.1 Types of Nylon Yarns

The diameter of the nylon yarn filament can be determined by the rate of delivery from the pump to the spinneret and the rate by which the yarn is drawn away from the spinneret. The size of yarn, which is measured in denier, can be determined by the diameter and the number of filaments in the yarn. Based upon this nylon can be divided as Nylon 6.6, Nylon 6.12, Nylon 4.6, Nylon 6, Nylon 12 and so on.

- Monofilament Yarns.
- Multifilament Yarns.
- Stretch Yarns.
- Textured Yarns.
- Spun Yarns.

3.4.2 Manufacturing Process of Nylon

U66IIU

Required amount of hexamethylene diamine and adipic acid are mixed to form nylon salt. This salt is dissolved in water at the spinning mill and heated to get a concentrated nylon salt solution. The solution is placed in an autoclave under high pressure and heat which results in giant chain links structure called "Linear Superpolymers". The solution is slowly passed over the autoclave and a molten nylon resin is deposited on wheel. This is sprayed with cold water to harden into opaque ribbon, flakes or pellets. On melting these pellets or flakes are passed through the spinneret to form nylon yarns (Figure 3.3).

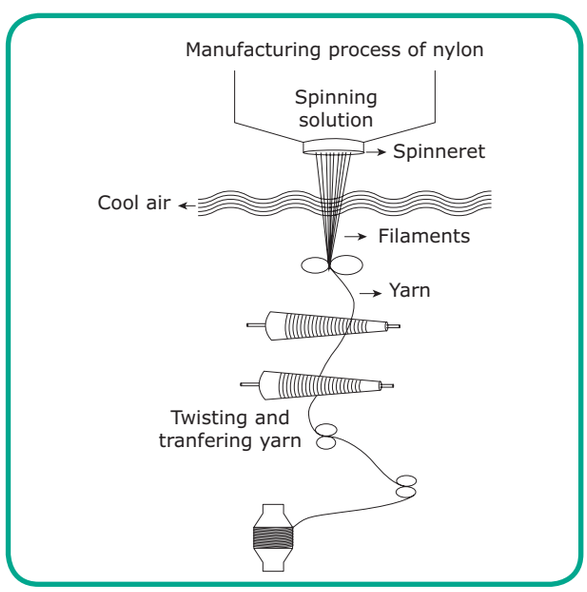


Figure 3.3 Manufacturing Process of Nylon

3.4.3 Properties of Nylon

Shape	Can be modified as per requirement.
Size	Thin long filament.
Density	1.1 g / ccm

TDD_Final_Chapter03.indd 28 21-02-2019 11.11.59 AM